

MedPRS: Scientific Paper Submission Recommendation System for Medical Research

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Abstract. A paper submission recommendation system aim to assist researchers in choosing suitable journals or conferences for their work. This research topic has been extensively studied during the last five years. The need for such systems is especially critical in the medical field, where the rapid expansion of biomedical literature makes selecting appropriate venues increasingly challenging. In this study, we propose three approaches developed on a newly constructed dataset comprising 1.2 million biomedical articles from 1,406 journals. The dataset is enriched with metadata such as titles, abstracts, keywords, journal Aims & Scope, and a newly introduced feature, Categories. By leveraging domain-specific transformer models, including BioBERT and BioMedBERT, our system achieves strong performance, with Top-1 to Top-10 accuracies ranging from 0.6865 to 0.9582.

Keywords: Paper submission · Recommendation system · Transformers
· Biomedical language models · Contrastive learning

1 Introduction

The rapid growth of scientific research has increased the number of journals and conferences, making it difficult - especially for early-career researchers and non-native English speakers - to identify suitable venues. Evaluating journal scope, indexing, and impact metrics often requires expertise many authors lack. Although recent submission recommendation systems have shown strong results [1–5], most focus on computer science and applied mathematics, leaving the biomedical domain largely unexplored. The biomedical field has also grown rapidly. For example, the number of Medicine - category journals listed by the SCImago Journal Rank (SJR)⁴ increased from 5,883 in 1999 to 7,743 in 2024, while PubMed Central (PMC)⁵ now hosts more than 11.1 million articles. This scale amplifies the need for domain-specific recommendation systems. Although

⁴ <https://www.scimagojr.com/journalrank.php>

⁵ <https://pubmed.ncbi.nlm.nih.gov>

several systems have attempted to support biomedical submission [6–9], most rely on general-purpose language models or insufficiently capture biomedical journal characteristics, leaving room for more specialized solutions.

To address this gap, we propose a framework that combines state-of-the-art recommendation architectures with domain-specific pre-trained language models such as BioBERT [10] and BioMedBERT [11]. We construct a large-scale dataset of 1,203,464 biomedical articles from 1,406 journals, enriched with titles, abstracts, keywords, Aims & Scope, and a new attribute, Categories, to better align articles with journal themes. Using these resources, our experiments demonstrate significant performance gains over existing approaches, providing a solid foundation for advancing biomedical paper submission recommendation.

2 Related Works

Early journal recommendation systems focused primarily on the computer science domain. Wang et al. [1] introduced one of the first models by extracting features from abstracts using Chi-square and TF-IDF, then applying logistic regression, achieving a top-3 accuracy of 61.37%. Son et al. [3] later improved performance by incorporating four attributes: title, abstract, keywords, and aims & scope - within a deep learning architecture, achieving 78.89% top-3 accuracy. Building on this, Son et al. [5] enhanced the framework by replacing FastText [12] with DistilBERT [13], pushing top-3 accuracy to 90.32%. More recently, Duc et al. [14] proposed a transformer-based recommendation approach utilizing supervised contrastive learning [15]. Their two-stage pipeline first fine-tunes a pre-trained language model to obtain more discriminative document embeddings, then applies the encoder to various attribute combinations, including a journal’s Aims & Scope. The results show competitive performance and further improvements over earlier architectures. In contrast, relatively few studies specifically target the biomedical domain [6–9]. Among them, Pubmender [8] remains one of the most prominent systems. Developed on a large dataset of 880,165 biomedical papers from 1,130 journals in PubMed Central, Pubmender employs pre-trained word2vec embeddings and CNNs to encode biomedical abstracts. Although effective at the time, achieving a top-3 accuracy of 71%, its performance is still considerably lower than more recent transformer-based systems in computer science, which surpass 90% [3, 5, 14]. This gap underscores the need for domain-adapted language models and richer biomedical-specific metadata in recommendation tasks. To bridge this gap, our study integrates transformer-based recommendation frameworks with biomedical pre-trained language models (BioBERT, BioMedBERT) and introduces additional journal attributes tailored to biomedical thematic structures, offering a more domain-sensitive and effective solution.

3 Dataset

In this study, we present a large-scale biomedical dataset comprising 1,203,464 scientific papers published in 1,406 journals between 2021 and 2024, collected

from PubMed. The dataset can be organized into two major groups: (1) scientific papers containing the title, abstract, and keywords, and (2) journal metadata including details, such as Aims & Scope, SJR rank, best rank, SJR index, best quartile, H-index, and journal categories. To ensure quality and relevance of the proposed method, we restricted our selection to journals ranked within the Top 1,500 across key biomedical disciplines - namely *Biochemistry, Genetics and Molecular Biology; Immunology and Microbiology; Medicine; Neuroscience; Nursing; Pharmacology, Toxicology and Pharmaceutics; Psychology; Dentistry; and Health Professions*.

Furthermore, we included only journals that published at least 100 PubMed-indexed articles between January 2021 and December 2024. To prepare the dataset, we applied a three-step preprocessing pipeline. First, we removed entries missing key textual fields (title, abstract, keywords) and discarded duplicates. Second, we standardized the text by removing URLs and non-alphabetic characters, splitting CamelCase tokens, lowercasing, removing stopwords (NLTK + biomedical-specific), and normalizing whitespace. Finally, journal titles from different sources were harmonized using a predefined mapping dictionary for consistent naming. This preprocessing ensures a high-quality, domain-aligned dataset suitable for developing and evaluating biomedical paper submission recommendation models. An example entry is shown in Table 1.

Table 1. A sample entry from our dataset

Scientific Paper	Journals
Title: A chemo mechanical constitutive model for muscle activation in bat wing skins. Abstract: Birds, bats and insects have evolved unique wing structures to achieve a wide range of flight capabilities. Insects have relatively stiff and passive wings, birds have a complex and hierarchical feathered structure and bats have an articulated skeletal system integrated with a highly stretchable skin ... Keywords: bat wing skin; chemo mechanical; constitutive modelling; skeletal muscle; viscoelastic.	Journal Name: Journal of the Royal Society Interface Aims and Scope: J. R. Soc. Interface welcomes articles of high-quality research at the interface of the physical and life sciences. It provides a high-quality forum to publish rapidly and interact across this boundary in two main ways: J. R. Soc. Interface publishes research applying chemistry, engineering, materials science, mathematics, and physics to the biological and medical sciences ... Categories: Biochemistry, Biophysics, Biotechnology

4 Methodology

In this section, we present three approaches applied to our biomedical dataset. These approaches form the basis of our proposed recommendation framework, allowing us to assess and improve model performance within the biomedical domain.

4.1 The First Method (Approach A)

In this work, we extend the approach of Huynh et al. [3] by introducing additional feature groups and employing a more domain-specific language model. Unlike the original study, which considers only two feature groups, our method explores three: (1) paper information alone, (2) paper information combined with the journal’s Aims & Scope, and (3) paper information enriched with both the Aims & Scope and journal categories. To better capture biomedical semantics, we train a specialized language model instead of relying on conventional embedding methods [16], as illustrated in Figure 1.

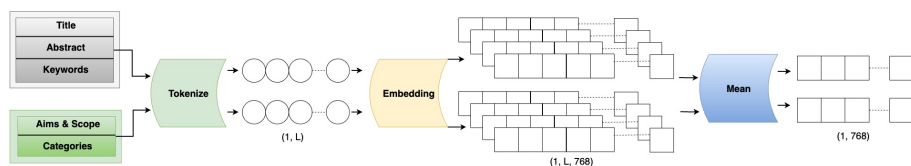


Fig. 1. Biomedical Language Model-Based Embedding. The model encodes a paper and a journal, tokenized to a maximum input length ($L=512$ tokens), into 768-dimensional contextual embeddings using a biomedical language model for domain-specific representation.

The first variant of our architecture consists of two parallel branches that encode paper and journal information using the biomedical language model. The resulting embeddings pass through a one-dimensional convolutional layer, global max pooling, a dense and dropout layers to extract salient features. A similarity extraction module (Figure 2) computes the similarity between the two representations, which is then refined and fed back as an additional feature to the journal branch. Finally, the outputs of both branches and the similarity feature are concatenated and passed through dense and dropout layers, followed by a softmax layer for classification, as shown in Figure 3.

4.2 The Second Method (Approach B)

In this approach, the architecture is implemented upon the framework that was presented by Son et al. [5]. Whereas the original model employed DistilBERT [13] as the embedding layer and utilized multi-scale one-dimensional convolutional layers to capture diverse input semantics, we extend this design by substituting DistilBERT with biomedical language models. Furthermore, we investigate two directions for containing journal categories: (1) treating them as a separate input feature, analogous to the Aims & Scope, or (2) merging them with the Aims & Scope before embedding and feeding them into the similarity module.

We preserve all remaining components of the initial architecture: the multi-kernel 1D convolutional feature extractor, global max pooling, fully connected layers,

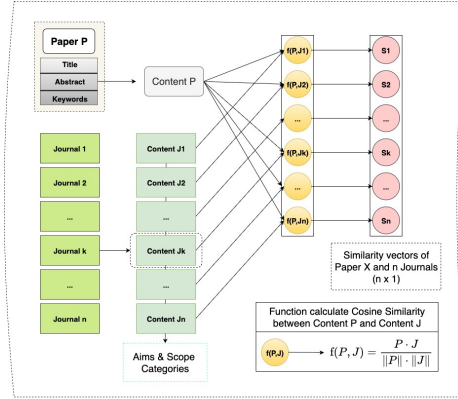


Fig. 2. Cosine similarity computation between article and journal embeddings

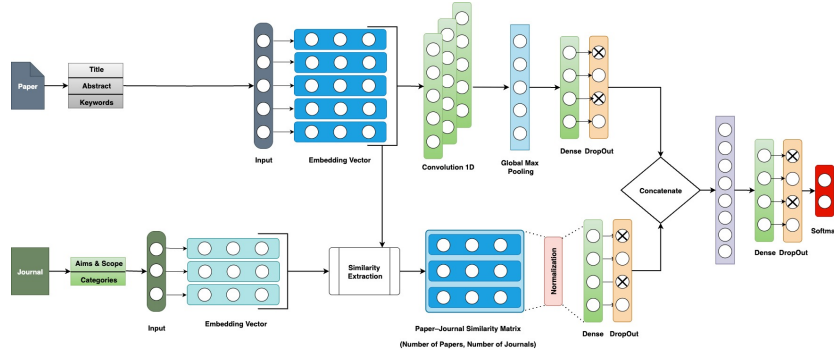


Fig. 3. Approach A Architecture. Approach A leverages five core attributes: the paper’s title, abstract, and keyword list, together with the journal’s Aims & Scope and assigned categories.

and the similarity scoring mechanism. The overall architecture is illustrated in Figure 4.

4.3 The Third Method (Approach C)

We propose a new architecture, utilizing biomedical language models as the backbone for document representation, and drawing inspiration from Le et al. [14]. It leverages a pretrained language model combined with a simple contrastive learning strategy to fine-tune all parameters and extends prior work by incorporating additional attributes specifically, journal categories and a more advanced language model architecture. Our approach includes two model variants. The first uses only paper-related information: the input is encoded by a biomedical language model and passed through a projection module (linear layer, ReLU activation, dropout). The resulting representation is fed into a final linear layer with a Softmax classifier (Figure 5). In the second variant, which integrates both

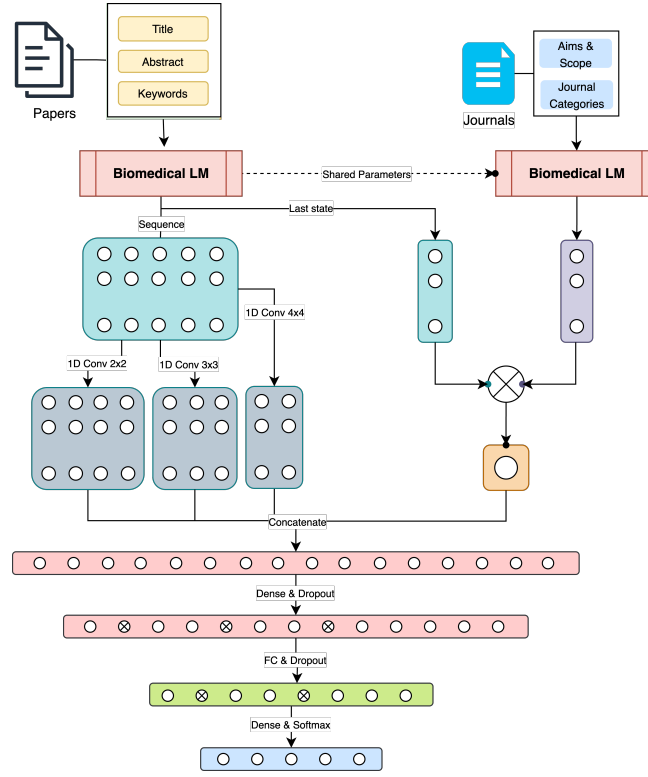


Fig. 4. Approach B Architecture. Approach B is an extension of the framework introduced by Huynh et al. [5], which adapts and refines their original design.

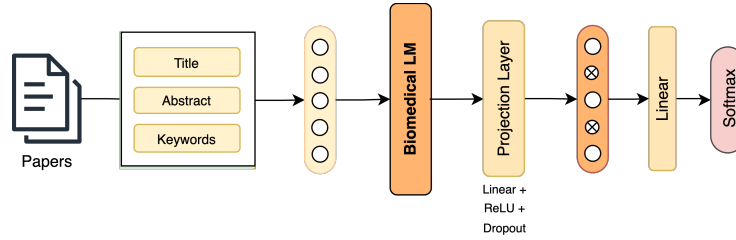


Fig. 5. The architecture of our approach C, which extends the SimCPSR approach [14], is designed for a model that utilizes only paper information as input.

paper and journal data, the architecture adopts a dual-branch structure. One branch encodes the paper using the same pipeline as above, while the other encodes the journal information using an identical setup. The two embeddings are fed into a Similarity Extraction module to compute a cosine similarity score, which is concatenated with the paper embedding to form a joint representation. This combined vector is then passed through a linear layer and Softmax classifier for final prediction (Figure 6).

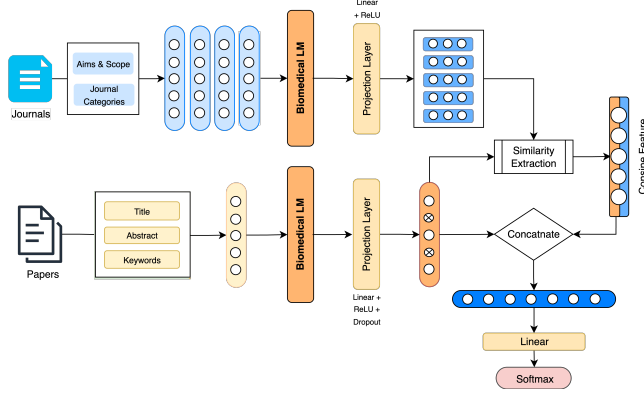


Fig. 6. The architecture of our approach C, which extends the SimCPSR approach [14], is designed for a model that utilizes both paper and journal information as input.

4.4 Evaluation Metrics

Our research employs the $Accuracy@K$ ($K = 1, 3, 5, 10$), as proposed in [3], to evaluate the performance of the paper submission recommendation system. The metric is defined by the following formula:

$$Accuracy@K = p_{top_K}(X, Y) = \frac{\sum_{x_i \in K, y_i \in K} P_{top_K}(x_i, y_i)}{|Y|} \quad (1)$$

where: X is the ground truth, and Y is the predicted one.

$$P_{top_K}(x_i, y_i) = \begin{cases} 1, & \text{if } y_i \text{ is in the Top } K \text{ list of recommended items by } x_i. \\ 0, & \text{otherwise} \end{cases}$$

5 Experiment

All experiments were conducted on a machine equipped with an Intel Core i9-9900K CPU, 64 GB of RAM, and an NVIDIA GeForce RTX 3090 GPU with 24 GB of VRAM.

5.1 Datasets

Our experiments were conducted on a biomedical dataset of 1,203,464 articles from 1,406 journals. We split the data into 70:10:20, corresponding to 842,424 articles for training, 120,346 for validation, and 240,694 for testing. All models were trained with Adam optimizer (learning rate $5e^{-5}$, batch size 64) for up to 10 epochs, using early stopping based on validation loss to prevent overfitting.

5.2 Experimental Results

In this section, we report experimental results on our biomedical dataset using three models: Approach A, Approach B, and Approach C. We evaluated

28 feature configurations combining paper content (Title(T), Abstract(A), Keywords(K)) with journal metadata (Aims & Scope(S), Categories(C)), and measured performance using Top-K accuracy (Top-1, Top-3, Top-5, Top-10). All models were implemented with BioBERT and BioMedBERT backbones.

To focus the analysis, Table 2 highlights the most informative feature groups: T, A, K, S, and C. Overall, the results show that augmenting the paper information with journal-level metadata - especially Aims & Scope and Categories - consistently improves classification performance across all three approaches. Among the three methods, Approach C achieves the best overall performance, followed by Approach B, with Approach A performing the lowest. Using BioBERT, Approach C improves Top-1,3,5, 10 accuracy by approximately 1% over Approach B and 3% over Approach A across most feature sets. Its highest Top-1 accuracy reaches 68.65% with the TAKSC feature combination, compared to 66.51% for Approach B and 63.53% for Approach A.

Table 2. Top-K (K=1,3,5,10) Accuracy of TAK, TAKC, TAKS, and TAKSC for three Approaches A, B, and C. The highest score within each approach is shown in bold, and the overall best performance across all models is highlighted in underlined bold red.

	Feature	BioBERT				BioMedBERT			
		Top 1	Top 3	Top 5	Top 10	Top 1	Top 3	Top 5	Top 10
Approach A	TAK	0.6105	0.8089	0.8716	0.9291	0.5455	0.7397	0.8094	0.8817
	TAKC	0.6220	0.8205	0.8817	0.9375	0.5555	0.7417	0.8114	0.8871
	TAKS	0.6188	0.8187	0.8813	0.9374	0.5663	0.7626	0.8303	0.8991
	TAKSC	0.6353	0.8317	0.8908	0.9432	0.5670	0.7637	0.8311	0.9010
	Feature	BioBERT				BioMedBERT			
		Top 1	Top 3	Top 5	Top 10	Top 1	Top 3	Top 5	Top 10
Approach B	TAK	0.6524	0.8470	0.9030	0.9519	0.6050	0.7991	0.8623	0.9219
	TAKC	0.6651	0.8529	0.9067	0.9530	0.6052	0.7973	0.8604	0.9207
	TAKS	0.6479	0.8431	0.9006	0.9489	0.6019	0.7965	0.8594	0.9201
	TAKSC	0.6651	0.8534	0.9065	0.9529	0.6033	0.7965	0.8598	0.9205
	Feature	Pre-trained BioBERT				Fine-tuned BioBERT			
		Top 1	Top 3	Top 5	Top 10	Top 1	Top 3	Top 5	Top 10
Approach C	TAK	0.6850	0.8683	0.9175	0.9578	0.6856	0.8630	0.9108	0.9511
	TAKC	0.6856	0.8693	0.9175	0.9582	0.6859	0.8643	0.9119	0.9517
	TAKS	0.6735	0.8588	0.9102	0.9530	0.6848	0.8628	0.9105	0.9511
	TAKSC	0.6754	0.8603	0.9105	0.9531	0.6865	0.8654	0.9124	0.9525

The positive impact of incorporating Categories (C) is evident for Approaches A and B. Adding C to the input consistently improves Top-K accuracy. For example, in Approach A, Top-1 accuracy increases from 0.6105 to 0.6220 when moving from TAK to TAKC and from 0.6188 to 0.6353 when moving from TAKS to TAKSC. Similarly, in Approach B, Top-1 accuracy rises from 0.6524 to 0.6651

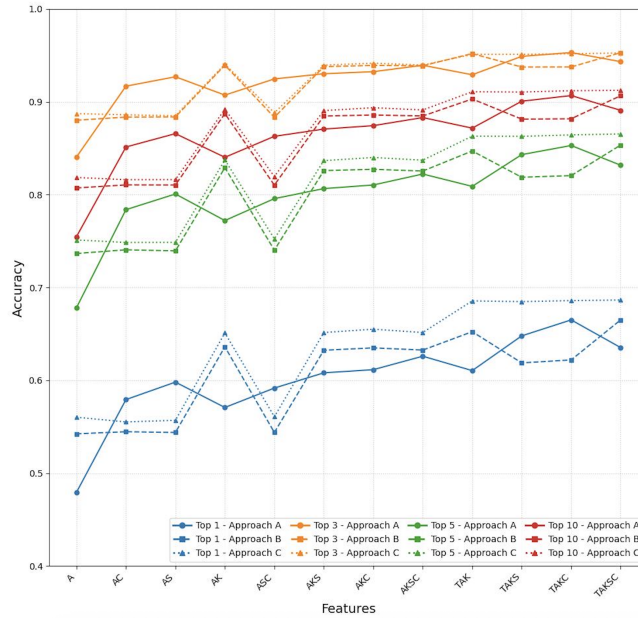


Fig. 7. Model Performance Comparison Across Approaches using BioBERT for Feature Combinations Containing the Abstract

(TAK to TAKC) and from 0.6479 to 0.6651 (TAKS to TAKSC). This trend is consistent for Top-3, Top-5, and Top-10 accuracy as well, confirming that Categories provide complementary signals to paper and Aims & Scope information. In contrast, Approach C shows minimal accuracy differences across feature combinations and between pretrained and fine-tuned BioBERT models. This suggests that the fine-tuning strategy does not provide substantial gains in this setting, possibly due to limited data for effective task-specific adaptation. In addition, when comparing the two language models, BioBERT generally outperforms BioMedBERT in Approaches A and B across most feature configurations. This indicates that BioBERT's pretraining objectives and vocabulary are better aligned with our dataset, whereas BioMedBERT does not provide additional advantages for these tasks.

Consistent with prior research, the Abstract feature continues to have a strong positive impact on model performance. All models achieve higher accuracy when the feature combinations include the Abstract, as shown in Figure 7 and Table 2. Furthermore, the results across the remaining feature groups reinforce the effectiveness of the proposed approaches in improving overall model performance. In summary, the experiments highlight three findings: (1) Approach C with BioBERT achieves the best overall performance compared to the other approaches, (2) Categories metadata significantly boost accuracy in biomedical paper submission recommendation, and (3) the Abstract feature continues to contribute strongly to model performance.

Table 3. Top-K Accuracy ($K = 1, 3, 5, 10$) of three approaches with BioBERT on the features group. The highest score across approaches for each attribute feature group is shown in bold.

Feature	Approach A				Approach B				Approach C			
	Top 1	Top 3	Top 5	Top 10	Top 1	Top 3	Top 5	Top 10	Top 1	Top 3	Top 5	Top 10
T	0.2892	0.4913	0.5877	0.7042	0.3460	0.5753	0.6773	0.7905	0.3524	0.5798	0.6796	0.7908
TC	0.5097	0.7457	0.8279	0.9058	0.3430	0.5718	0.6729	0.7872	0.3518	0.5790	0.6792	0.7892
TS	0.5428	0.7744	0.8501	0.9206	0.3441	0.5719	0.6730	0.7865	0.3498	0.5775	0.6772	0.7890
TSC	0.5465	0.7778	0.8533	0.9227	0.3429	0.5707	0.6719	0.7877	0.3503	0.5770	0.6768	0.7885
K	0.2639	0.4393	0.5217	0.6215	0.2959	0.4793	0.5644	0.6621	0.2988	0.4854	0.5691	0.6644
KC	0.4962	0.7302	0.8155	0.8972	0.2929	0.4793	0.5647	0.6622	0.2971	0.4833	0.5675	0.6635
KS	0.5274	0.7587	0.8373	0.9120	0.2937	0.4787	0.5635	0.6611	0.2985	0.4831	0.5676	0.6634
KSC	0.5362	0.7664	0.8446	0.9168	0.2933	0.4790	0.5643	0.6618	0.2979	0.4834	0.5686	0.6645
A	0.4794	0.6782	0.7547	0.8405	0.5424	0.7366	0.8072	0.8801	0.5603	0.7512	0.8185	0.8872
AC	0.5793	0.7838	0.8513	0.9168	0.5447	0.7406	0.8105	0.8835	0.5552	0.7485	0.8161	0.8860
AS	0.5981	0.8007	0.8657	0.9269	0.5439	0.7394	0.8104	0.8837	0.5570	0.7486	0.8162	0.8849
ASC	0.5917	0.7957	0.8628	0.9246	0.5439	0.7402	0.8103	0.8833	0.5608	0.7525	0.8196	0.8881
TK	0.4369	0.6702	0.7600	0.8537	0.5058	0.7441	0.8273	0.9062	0.5080	0.7445	0.8269	0.9047
TKC	0.5585	0.7880	0.8626	0.9287	0.5000	0.7406	0.8247	0.9041	0.5119	0.7500	0.8309	0.9067
TKS	0.5831	0.8063	0.8768	0.9375	0.5103	0.7394	0.8240	0.9041	0.5138	0.7492	0.8298	0.9063
TKSC	0.5826	0.8079	0.8774	0.9382	0.4969	0.7368	0.8216	0.9022	0.5119	0.7485	0.8297	0.9062
TA	0.5333	0.7312	0.8023	0.8784	0.6009	0.7958	0.8592	0.9200	0.6130	0.8023	0.8600	0.9173
TAC	0.6049	0.8069	0.8717	0.9306	0.5952	0.7905	0.8538	0.9164	0.6136	0.8038	0.8635	0.9199
TAS	0.6223	0.8213	0.8830	0.9381	0.5961	0.7912	0.8549	0.9168	0.6157	0.8053	0.8628	0.9187
TASC	0.6197	0.8201	0.8826	0.9382	0.5967	0.7916	0.8561	0.9184	0.6139	0.8031	0.8615	0.9180
AK	0.5708	0.7721	0.8402	0.9072	0.6361	0.8291	0.8872	0.9394	0.6488	0.8373	0.8920	0.9405
AKC	0.6115	0.8104	0.8743	0.9324	0.6350	0.8273	0.8858	0.9392	0.6551	0.8400	0.8936	0.9416
AKS	0.6081	0.8065	0.8706	0.9302	0.6324	0.8259	0.8846	0.9378	0.6515	0.8366	0.8905	0.9394
AKSC	0.6260	0.8222	0.8830	0.9393	0.6326	0.8254	0.8847	0.9388	0.6515	0.8371	0.8912	0.9395

6 Conclusion and Future Work

In conclusion, we proposed and evaluated enhanced architectures for biomedical paper submission recommendation by integrating domain-specific language models (BioBERT, BioMedBERT) with journal metadata such as Aims & Scope and Categories. Experiments on a large-scale biomedical dataset across 28 feature configurations show that leveraging domain adaptation and journal metadata significantly improves recommendation accuracy, highlighting the importance of combining domain knowledge with structured journal information for better paper-journal matching.

In the future, we aim to enhance recommendation models by exploring additional large language models and applying retrieval-augmented generation (RAG) for paper submission, adopting more advanced deep learning architectures, and investigating other contextual metadata to refine journal-paper matching.

References

1. Donghui Wang, Yanchun Liang, Dong Xu, Xiaoyue Feng, and Renchu Guan. A content-based recommender system for computer science publications. *Knowledge-based systems*, 157:1–9, 2018.

2. Dinh V Cuong, Dac H Nguyen, Son Huynh, Phong Huynh, Cathal Gurrin, Minh-Son Dao, Duc-Tien Dang-Nguyen, and Binh T Nguyen. A framework for paper submission recommendation system. In *Proceedings of the 2020 international conference on multimedia retrieval*, pages 393–396, 2020.
3. Son T Huynh, Nhi Dang, Phong T Huynh, Dac H Nguyen, and Binh T Nguyen. A fusion approach for paper submission recommendation system. In *International Conference on Industrial, Engineering and Other Applications of Applied Intelligent Systems*, pages 72–83. Springer, 2021.
4. Dac Huu Nguyen, Son Thanh Huynh, Cuong Viet Dinh, Phong Tan Huynh, and Binh Thanh Nguyen. Psrmte: Paper submission recommendation using mixtures of transformer. *Expert Systems with Applications*, 202:117096, 2022.
5. Son T Huynh, Nhi Dang, Dac H Nguyen, Phong T Huynh, and Binh T Nguyen. Fpsrs: a fusion approach for paper submission recommendation system. *Applied Intelligence*, 53(8):8614–8630, 2023.
6. Martijn J Schuemie and Jan A Kors. Jane: suggesting journals, finding experts. *Bioinformatics*, 24(5):727–728, 2008.
7. Özge Kart, Alexandre Mestiashvili, Kurt Lachmann, Richard Kwasnicki, and Michael Schroeder. Emati: a recommender system for biomedical literature based on supervised learning. *Database*, 2022:baac104, 2022.
8. Xiaoyue Feng, Hao Zhang, Yijie Ren, Penghui Shang, Yi Zhu, Yanchun Liang, Renchu Guan, and Dong Xu. The deep learning-based recommender system “pub-mender” for choosing a biomedical publication venue: Development and validation study. *Journal of medical Internet research*, 21(5):e12957, 2019.
9. Takashi Yoneya and Hiroshi Mamitsuka. Pure: a pubmed article recommendation system based on content-based filtering. *Genome informatics*, 18:267–276, 2007.
10. Jinhyuk Lee, Wonjin Yoon, Sungdong Kim, Donghyeon Kim, Sunkyu Kim, Chan Ho So, and Jaewoo Kang. Biobert: a pre-trained biomedical language representation model for biomedical text mining. *Bioinformatics*, 36(4):1234–1240, 2020.
11. Yu Gu, Robert Tinn, Hao Cheng, Michael Lucas, Naoto Usuyama, Xiaodong Liu, Tristan Naumann, Jianfeng Gao, and Hoifung Poon. Domain-specific language model pretraining for biomedical natural language processing. *ACM Transactions on Computing for Healthcare (HEALTH)*, 3(1):1–23, 2021.
12. Piotr Bojanowski, Edouard Grave, Armand Joulin, and Tomas Mikolov. Enriching word vectors with subword information. *Transactions of the Association for Computational Linguistics*, 5:135–146, 2017.
13. Victor Sanh, Lysandre Debut, Julien Chaumond, and Thomas Wolf. Distilbert, a distilled version of bert: smaller, faster, cheaper and lighter. *arXiv preprint arXiv:1910.01108*, 2019.
14. Duc H Le, Tram T Doan, Son T Huynh, and Binh T Nguyen. Simcpsr: Simple contrastive learning for paper submission recommendation system. In *Asian Conference on Intelligent Information and Database Systems*, pages 51–63. Springer, 2022.
15. Tianyu Gao, Xingcheng Yao, and Danqi Chen. Simcse: Simple contrastive learning of sentence embeddings. *arXiv preprint arXiv:2104.08821*, 2021.
16. Piotr Bojanowski, Edouard Grave, Armand Joulin, and Tomas Mikolov. Enriching word vectors with subword information. *Transactions of the association for computational linguistics*, 5:135–146, 2017.